

Link to github

https://github.com/pmozil/system-arch-homework/tree/micro_basics

Architecture

The system consists of three microservices:

- **facade-service**: Acts as an API gateway, accepting HTTP POST/GET requests from clients
- **logging-service**: Stores messages in memory and provides access to them via gRPC
- **messages-service**: A stub service that returns a static message via HTTP

Communication Flow

POST Request Flow:

1. Client sends POST request to facade-service with a message
2. facade-service generates a UUID for the message
3. facade-service forwards the message to logging-service via gRPC
4. logging-service stores the message in memory
5. Response is returned to the client with the UUID

GET Request Flow:

1. Client sends GET request to facade-service
2. facade-service requests messages from logging-service via gRPC
3. facade-service requests message from messages-service via HTTP
4. facade-service concatenates responses and returns to client

Prerequisites

- Rust

Project Structure

```
|-- Cargo.toml
|-- proto
|   -- logging.proto
|---src
|   |-- facade-service.rs
|   |-- logging-service.rs
|   |-- messages-service.rs
```

Installation

1. Clone the repository

2. Ensure you have the Protocol Buffers compiler installed
3. Build the project:

```
cargo build
```

Running the Services

Start each service in a separate terminal:

```
# Terminal 1 - Start logging-service  
cargo run --bin facade-service
```

```
# Terminal 2 - Start facade-service  
cargo run --bin logging-service
```

```
# Terminal 3 - Start messages-service  
cargo run --bin messages-service
```

Service addresses: - facade-service: <http://localhost:3000> - logging-service: [http://\[::1\]:50051](http://[::1]:50051) (gRPC) - messages-service: <http://localhost:3002>

Testing the Services

You can test the services using curl:

```
# Send a message  
curl -X POST http://localhost:3000/message \\\n  -H "Content-Type: application/json" \\\n  -d '{"msg":"Hello World"}'
```

```
# Get all messages  
curl http://localhost:3000/messages
```

Results

My commands:

```
[petro@shire sa-homework]$ curl -X POST http://localhost:3000/message -H \\\n  "Content-Type: application/json" -d '{"msg":"Hello"}'  
{\"id\":\"b849ed06-1526-4491-bcc5-53f6d29453cf\",\"msg\":\"Hello\"}  
[petro@shire sa-homework]$ curl -X POST http://localhost:3000/message -H \\\n  "Content-Type: application/json" -d '{"msg\":\"Hello1\"}'  
{\"id\":\"f45a556c-126f-4d4a-a289-39232e08578e\",\"msg\":\"Hello1\"}  
[petro@shire sa-homework]$ curl -X POST http://localhost:3000/message -H \\\n  "Content-Type: application/json" -d '{"msg\":\"Hello2\"}'  
{\"id\":\"146139c0-2f1f-4d89-9b43-5f4598b30960\",\"msg\":\"Hello2\"}  
[petro@shire sa-homework]$ curl -X POST http://localhost:3000/message -H \\\n  "Content-Type: application/json" -d '{"msg\":\"Hello3\"}'
```

```
[petro@shire sa-homework]$ curl http://localhost:3000/messages
Hello1
Hello2
Hello
Hello3
```

Log of facade:

```
[petro@shire sa-homework]$ ./target/debug/logging-service
Logging service listening on [::1]:50051
Received message: LogRequest { id: "b849ed06-1526-4491-bcc5-53f6d29453cf", msg: "Hello" }
Received message: LogRequest { id: "f45a556c-126f-4d4a-a289-39232e08578e", msg: "Hello1" }
Received message: LogRequest { id: "146139c0-2f1f-4d89-9b43-5f4598b30960", msg: "Hello2" }
Received message: LogRequest { id: "29925212-808e-478c-9579-ade0e304de0e", msg: "Hello3" }
```

Log of logger:

```
[petro@shire sa-homework]$ ./target/debug/facade-service
Facade service listening on 127.0.0.1:3000`
```

Log of message

```
[petro@shire sa-homework]$ ./target/debug/message-service
Messages service listening on 127.0.0.1:3002
```